

A large industrial steam turbine is shown in a factory setting. The turbine is composed of several large, circular, metallic components with many small, curved blades. The central part of the turbine is a large, polished metal cylinder. The background shows the interior of a factory with various structural elements and a person standing near the turbine for scale.

STEAM TURBINES

STEAM TURBINES

A steam turbine is a prime mover in which the potential energy of the steam is transformed into kinetic energy and later in its turn is transformed into the mechanical energy of rotation of the turbine shaft.

APPLICATIONS OF STEAM TURBINES

- Power Generation
- Transport

CLASSIFICATION OF STEAM TURBINES

According to the action of steam:

Impulse turbine:

- In impulse turbine, steam coming out through a fixed nozzle at a very high velocity strikes the blades fixed on the periphery of a rotor.
- The blades change the direction of steam flow without changing its pressure.
- The force due to change of momentum causes the rotation of the turbine shaft.
- Ex: De-Laval, Curtis and Rateau Turbines

Reaction turbine:

- In reaction turbine, steam expands both in
- fixed and moving blades continuously as the steam passes over
- them. The pressure drop occurs continuously over both moving
- and fixed blades.

Combination of impulse and reaction turbine

CLASSIFICATION OF STEAM TURBINES

According to the number of pressure stages:

Single stage turbines:

These turbines are mostly used for driving centrifugal compressors, blowers and other similar machinery.

Multistage Impulse and Reaction turbines:

They are made in a wide range of power capacities varying from small to large.

According to the type of steam flow:

Axial turbines:

In these turbines, steam flows in a direction parallel to the axis of the turbine rotor.

Radial turbines:

In these turbines, steam flows in a direction perpendicular to the axis of the turbine, one or more low pressure stages are made axial.

CLASSIFICATION OF STEAM TURBINES

According to the number of shafts:

- Single shaft turbines
- Multi-shaft turbines

According to the method of governing:

- **Turbines with throttle governing:** In these turbines, fresh steam enter through one or more (depending on the power developed) simultaneously operated throttle valves.
- **Turbines with nozzle governing:** In these turbines, fresh steam enters through one or more consecutively opening regulators.
- **Turbines with by-pass governing:** In these turbines, the steam besides being fed to the first stage is also directly fed to one, two or even three intermediate stages of the turbine.

CLASSIFICATION OF STEAM TURBINES

According to the heat drop process:

Condensing turbines with generators:

- In these turbines, steam at a pressure less than the atmospheric is directed to the condenser. The steam is also extracted from intermediate stages for feed water heating). The latent heat of exhaust steam during the process of condensation is completely lost in these turbines.

Condensing turbines with one or more intermediate stage extractions:

- In these turbines, the steam is extracted from intermediate stages for industrial heating purposes.

Back pressure turbines:

- In these turbines, the exhaust steam is utilized for industrial or heating purposes. Turbines with deteriorated vacuum can also be used in which exhaust steam may be used for heating and process purposes.

Topping turbines: In these turbines, the exhaust steam is utilized in medium and low pressure condensing turbines. These turbines operate at high initial conditions of steam pressure and temperature, and are mostly used during extension of power station capacities, with a view to obtain better efficiencies.

CLASSIFICATION OF STEAM TURBINES

*According to the steam conditions
at inlet to turbine:*

Low pressure turbines:

These turbines use steam at a pressure of 1.2 ata to 2 ata.

Medium pressure turbines:

These turbines use steam up to a pressure of 40 ata.

High pressure turbines:

These turbines use steam at a pressure above 40 ata.

Very high pressure turbines:

These turbines use steam at a pressure of 170 ata and higher and temperatures of 550°C and higher.

Supercritical pressure turbines:

These turbines use steam at a pressure of 225 ata and higher.

CLASSIFICATION OF STEAM TURBINES

According to their usage in industry:

Stationary turbines with constant speed of rotation:

These turbines are primarily used for driving alternators.

Stationary turbines with variable speed of rotation:

These turbines are meant for driving turbo-blowers, air circulators, pumps, etc.

Non-stationary turbines with variable speed:

These turbines are usually employed in steamers, ships and railway locomotives.

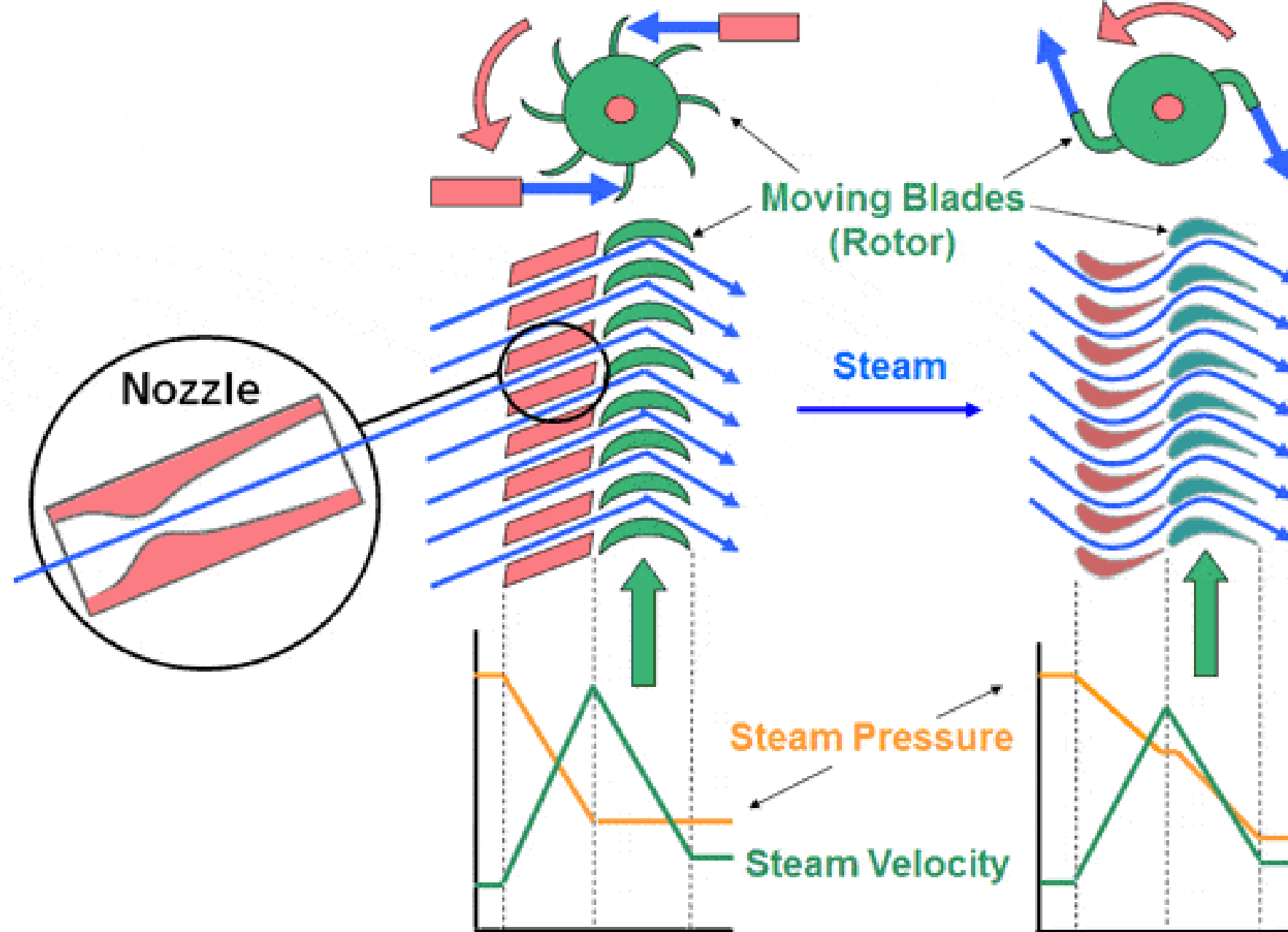
ADVANTAGES OF STEAM TURBINES OVER STEAM ENGINES

- The thermal efficiency is much higher.
- As there is no reciprocating parts, perfect balancing is possible and therefore heavy foundation is not required.
- Higher and greater range of speed is possible.
- The lubrication is very simple as there are no rubbing parts.
- The power generation is at uniform rate & hence no flywheel is required.
- The steam consumption rate is lesser.
- More compact and require less attention during operation.
- More suitable for large power plants.
- Less maintenance cost as construction and operation is highly simplified due to absence of parts like piston, piston rod, cross head, connecting rod.
- Considerable overloads can be carried at the expense of slight reduction in overall efficiency.

S.no	Impulse Turbine	Reaction Turbine
1.	In impulse turbine the steam flows through the nozzle and strike on the moving blades.	In the reaction turbine, first the steam flows through the guide mechanism and then flows through the moving blades.
2.	Steam strikes on the buckets with kinetic energy.	The steam glides over the moving blades with both pressure and kinetic energy.
3.	During the flow of steam through moving blades, its pressure remains constant.	During the flow of steam through moving blades its pressure reduces.
4.	The steam may or may not be admitted to the whole circumference.	The steam must be admitted over the whole circumference.
5.	The blades of impulse turbine are symmetrical.	The blades of reaction turbine are not symmetrical.
6.	While gliding over the blades the relative velocity of steam remains constant.	In reaction turbine, while gliding over the blades the relative velocity of steam increases.
7.	For the same power developed, the number of stages required is less.	For the same power developed, the number of stages required is more.
8.	The direction of steam flow is radial to the direction of turbine wheel.	The direction of steam flow is radial and axial to the turbine wheel.
9.	It requires less maintenance work.	It requires more maintenance work.
10.	It is suitable for low discharge.	It is suitable for medium and high discharge.
11.	Pelton Wheel is the example of impulse turbine.	Francis turbine, Kaplan turbine etc. are the examples of reaction turbine.

Impulse Turbine

Reaction Turbine



IMPULSE PRINCIPLE

- The impulse turbine consists basically of a rotor mounted on a shaft that is free to rotate in a set of bearings.
- The outer rim of the rotor carries a set of curved blades, and the whole assembly is enclosed in an airtight case.
- Nozzles direct steam against the blades and turn the rotor. The energy to rotate an impulse turbine is derived from the kinetic energy of the steam flowing through the nozzles.
- The term impulse means that the force that turns the turbine comes from the impact of the steam on the blades.

IMPULSE PRINCIPLE

- The toy pinwheel can be used to study some of the basic principles of turbines. When we blow on the rim of the wheel, it spins rapidly. The harder we blow, the faster it turns.
- The steam turbine operates on the same principle, except it uses the kinetic energy from the steam as it leaves a steam nozzle rather than air.
- Steam nozzles are located at the turbine inlet. As the steam passes through a steam nozzle, potential energy is converted to kinetic energy.
- This steam is directed towards the turbine blades and turns the rotor. The velocity of the steam is reduced in passing over the blades.
- Some of its kinetic energy has been transferred to the blades to turn the rotor.
- Impulse turbines may be used to drive forced draft blowers, pumps, and main propulsion turbines.

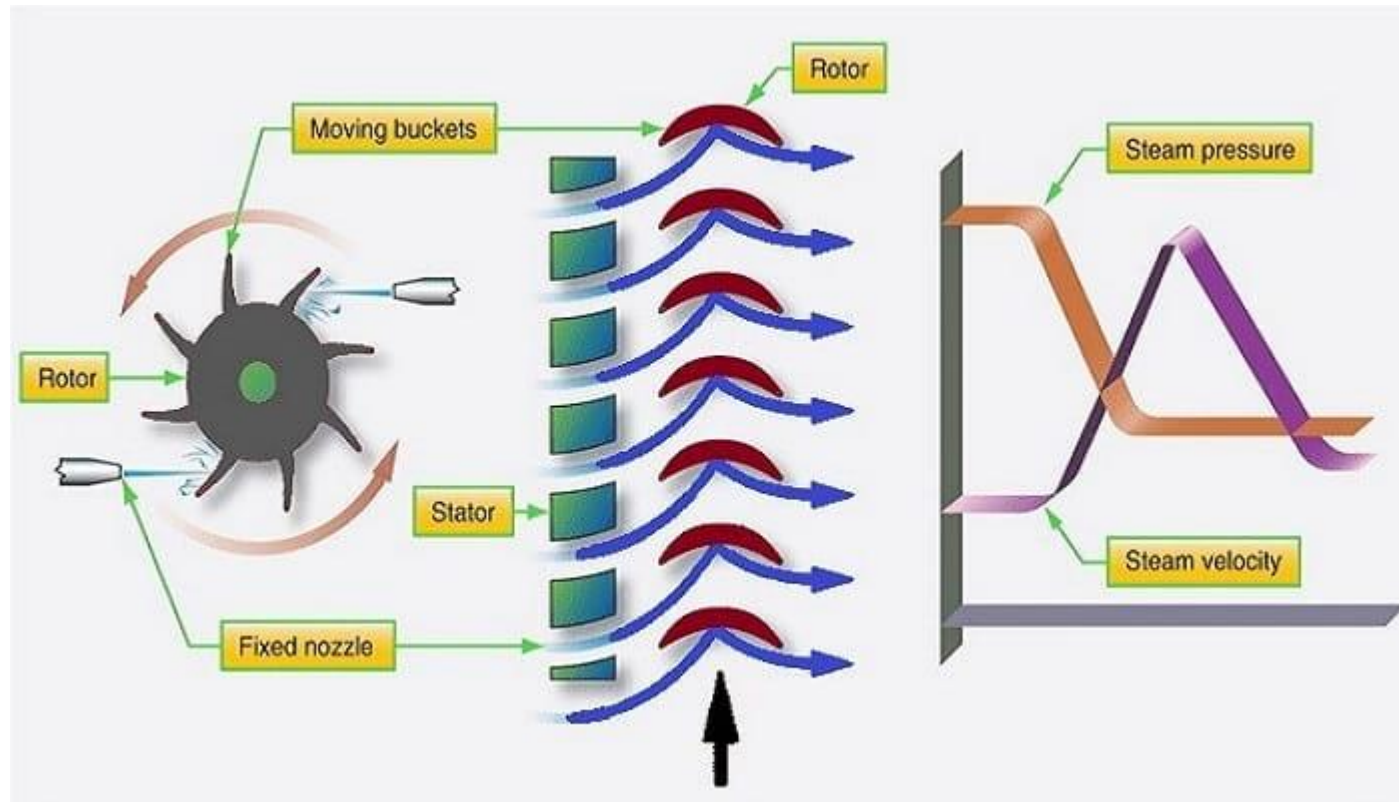
IMPULSE PRINCIPLE

- In impulse turbine, steam coming out through a fixed nozzle at a very high velocity strikes the blades fixed on the periphery of a rotor.
- The blades change the direction of steam flow without changing its pressure.
- The force due to change of momentum causes the rotation of the turbine shaft.
- Examples: De-Laval, Curtis and Rateau turbines.

CONSTRUCTION & WORKING PRINCIPLE OF IMPULSE PRINCIPLE

- It primarily consists of a nozzle or a set of nozzles, a rotor mounted on a shaft, one set of moving blades attached to the rotor and a casing.
- A simple impulse turbine is also called De-Laval turbine, after the name of its inventor
- This turbine is called **simple** impulse turbine since the expansion of the steam takes place in one set of nozzles.

CONSTRUCTION & WORKING PRINCIPLE OF IMPULSE PRINCIPLE



IMPULSE TURBINE

- The uppermost portion of the diagram shows a longitudinal section through the upper half of the turbine.
- The middle portion shows the actual shape of the nozzle and blading.
- The bottom portion shows the variation of absolute velocity and absolute pressure during the flow of steam through passage of nozzles and blades.
- The expansion of steam from its initial pressure (steam chest pressure) to final pressure (condenser pressure) takes place in one set of nozzles.
- Due to high drop in pressure in the nozzles, the velocity of steam in the nozzles increases.

IMPULSE TURBINE

- The steam leaves the nozzle with a very high velocity and strikes the blades of the turbine mounted on a wheel with this high velocity.
- The loss of energy due to this higher exit velocity is commonly known as carry over loss (or) leaving loss.
- The pressure of the steam when it moves over the blades remains constant but the velocity decreases.
- Since all the KE is to be absorbed by one ring of the moving blades only, the velocity of wheel is too high (varying from 25000 to 30000 RPM).
- However, this wheel or rotor speed can be reduced by adopting the method of compounding of turbines.

DISADVANTAGES OF IMPULSE TURBINE

- Since all the KE of the high velocity steam has to be absorbed in only one ring of moving blades, the velocity of the turbine is too high i.e. up to 30,000 RPM for practical purposes.
- The velocity of the steam at exit is sufficiently high which means that there is a considerable loss of KE.

COMPOUNDING OF TURBINE

- In a simple turbine the energy of steam/ water extracted in single stage.
- The main problem with the single stage operation is that the large portion of energy of may get lose due to higher exit velocity and pressure.
- This loss is known as carry over loss/ leaving loss.
- The one stage operation cause an extremely high speed of rotor which cause design complication due to high vibrations.
- This eventually leads the damage of structure.

COMPOUNDING OF TURBINE

- In order to avoid these drawbacks and reduce the speed of rotor, the energy of fluid extracted in number of stages (in multiple stages) rather than a single stage.
- This method is known as compounding of turbine.
- Sets of blades and nozzle are arranged in such a way that the pressure energy and kinetic energy are absorbed in stages as the working fluid flow over the blades.
- The multiple rotor is then keyed to common shaft.

TYPES OF COMPOUNDING OF TURBINE

According to the arrangement and of blades and nozzle, and the way the energy extracted, the compounding is divided into

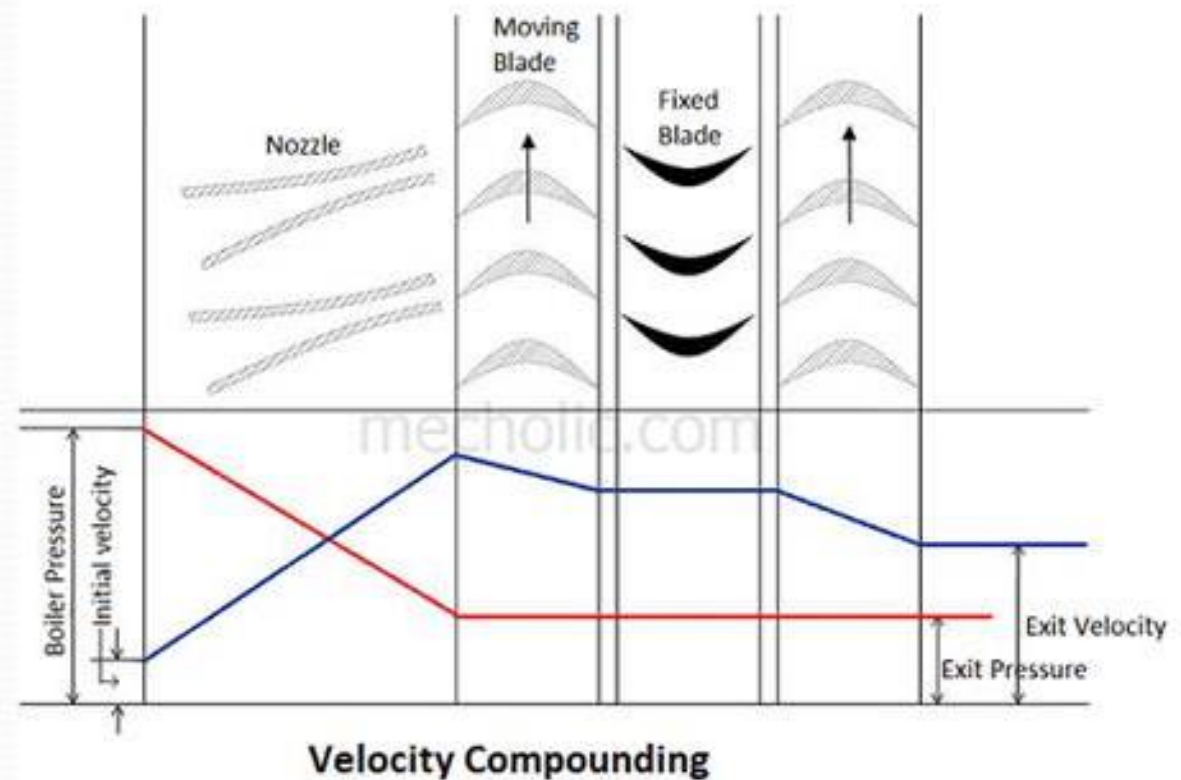
Velocity compounding: Kinetic energy (velocity) of Working fluid (steam) is extracted, and convert it to useful work in multiple stages. The pressure drop (steam expand) occurs at first nozzle only.

Pressure compounding: Steam expansion (pressure drop) occur in multiple stages.

Pressure-velocity compounding: it is a combination of both velocity compounding and pressure compounding. Both velocity and pressure energy extracted in multiple stages.

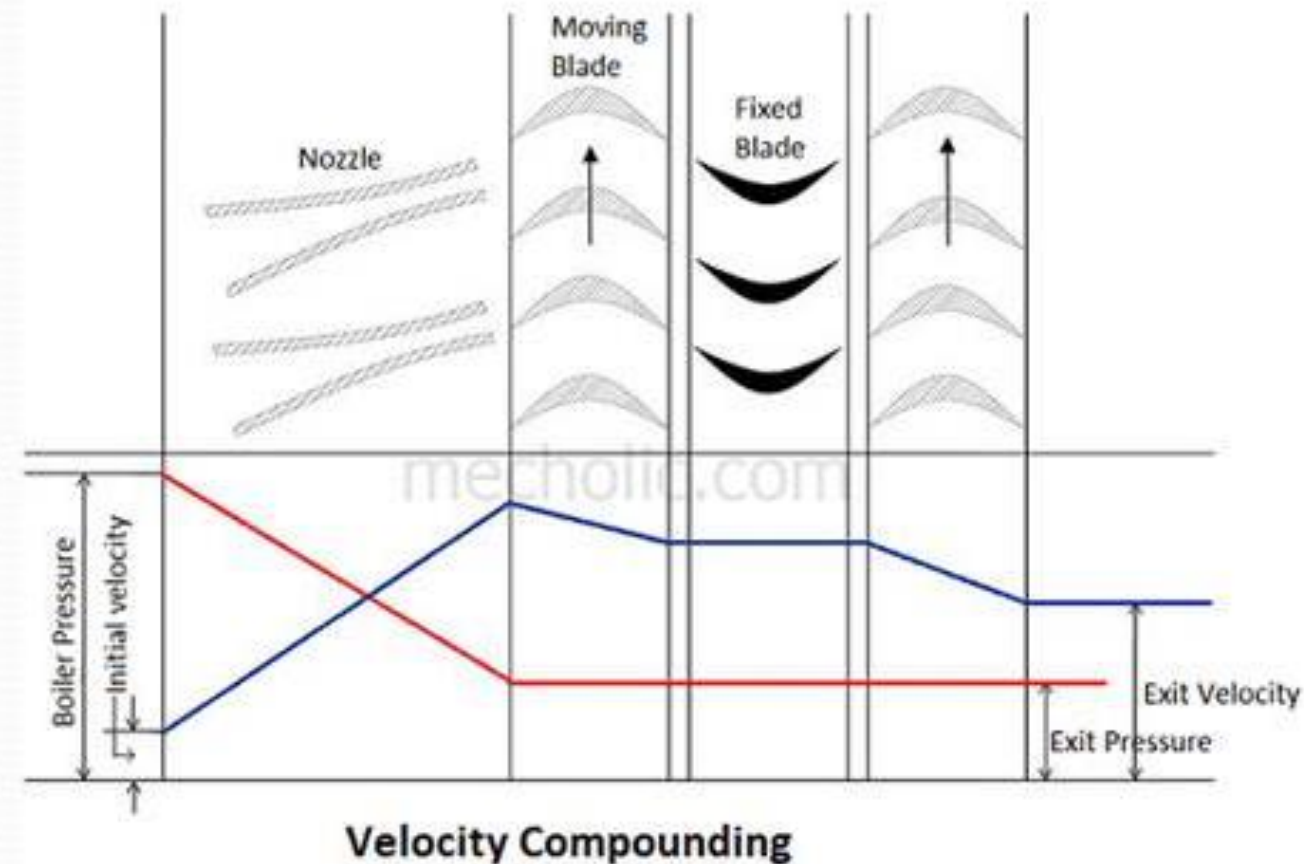
VELOCITY-COMPOUNDED IMPULSE TURBINE

- Velocity drop is arranged in many small drops through many moving blades instead of single row of moving blades.
- High pressure steam is expanded in nozzle which is then transferred to first set of moving blades where steam loses part of its kinetic energy.



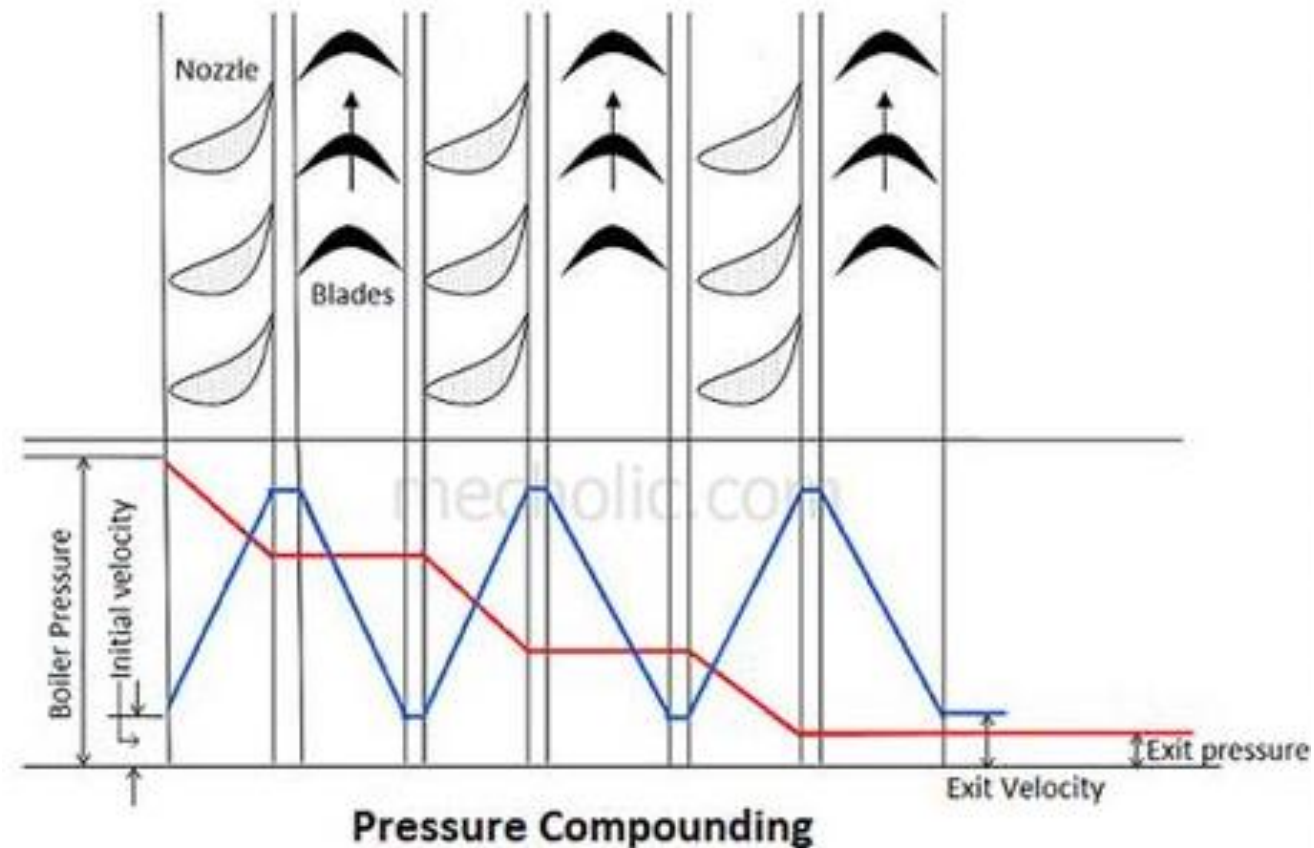
VELOCITY-COMPOUNDED IMPULSE TURBINE

- Fixed blade are guide blades that guide the steam to succeeding row of fixed blade which further reduces its velocity.



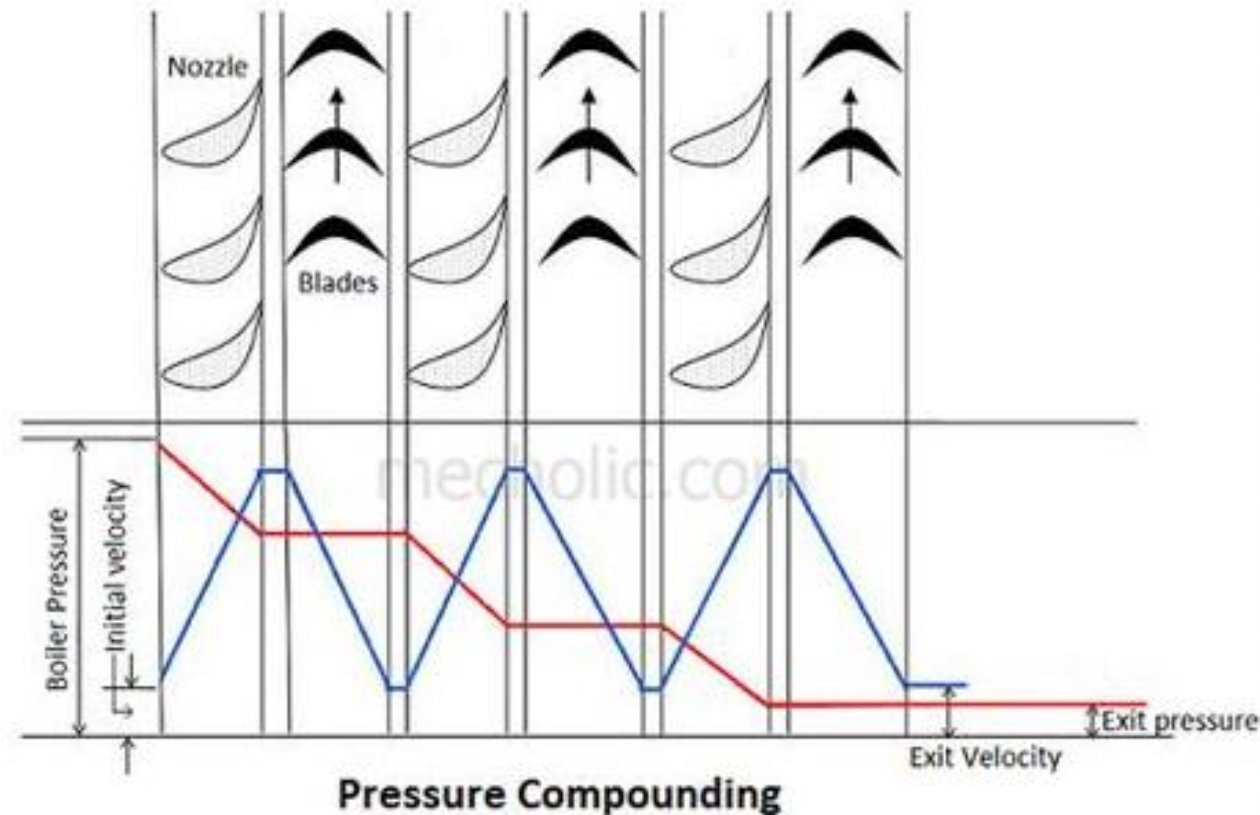
PRESSURE-COMPOUNDED IMPULSE TURBINE

- In this pressure drop from chest pressure to the condenser pressure is split up into smaller pressure drop across several stages of impulse turbine.



PRESSURE-COMPOUNDED IMPULSE TURBINE

- It consists of alternate rings of nozzles and turbine blades. The nozzles are fitted to the casing and the blades are keyed to the turbine shaft.
- In this type of compounding the steam is expanded in a number of stages, instead of just one (nozzle) in the velocity compounding.



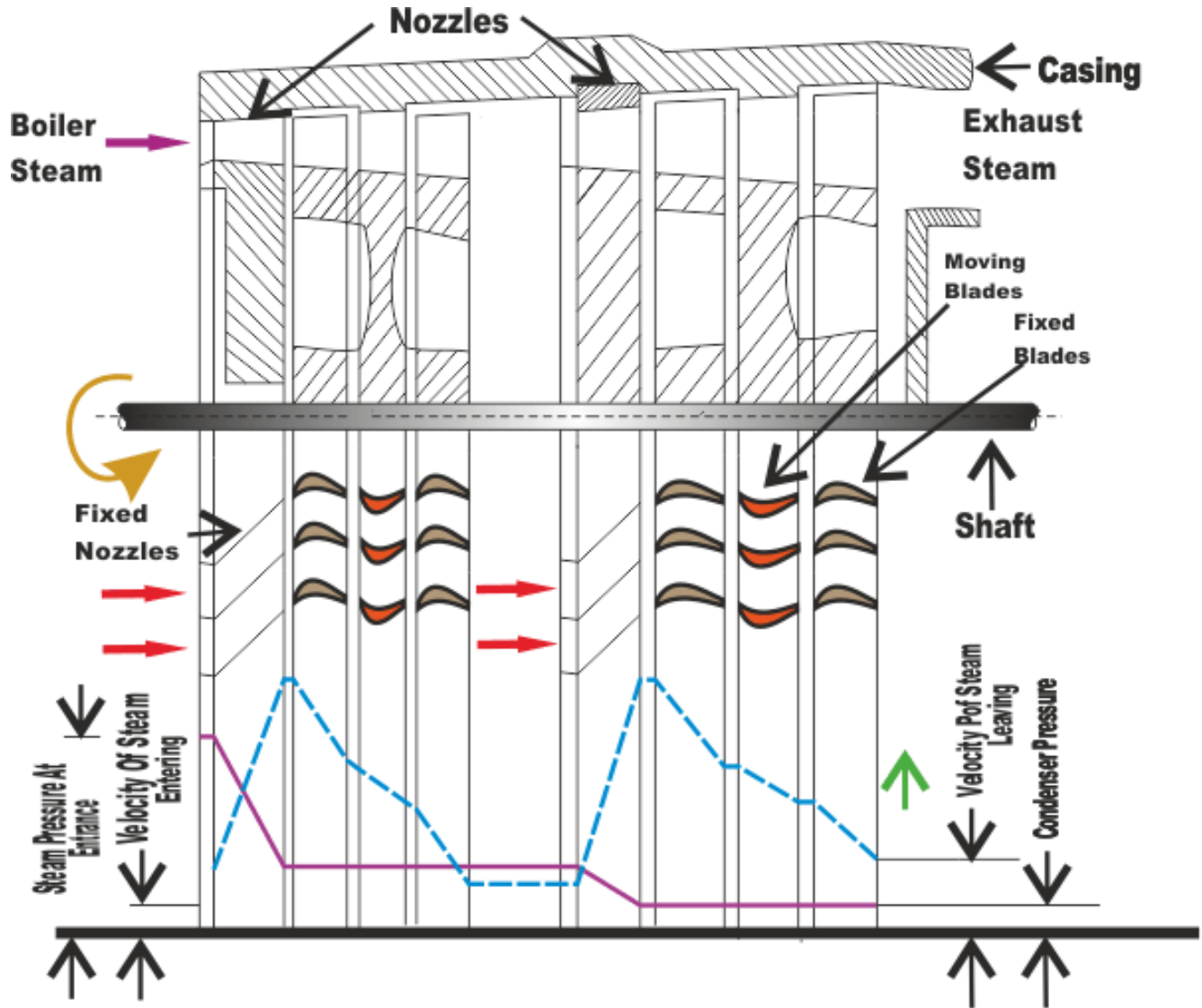
PRESSURE-VELOCITY COMPOUNDED IMPULSE TURBINE

- This turbine is the combination of pressure and velocity compounding.
- It consist of set of nozzles each consist of moving and fixed blades. Pressure compounding occurs in set of nozzles while velocity compounding occurs in moving and fixed blades.

PRESSURE-VELOCITY COMPOUNDED IMPULSE TURBINE

- In first set of nozzles, slight decrease in pressure occurs while the kinetic energy increases. There is no pressure drop in two rows of moving blade. However, velocity drops in moving blades.
- In second set of nozzles, the remaining pressure drop takes place, but velocity increases and, the drop in velocity takes place in moving blades of the second rotor

- In first set of nozzles, slight decrease in pressure occurs while the kinetic energy increases. There is no pressure drop in two rows of moving blade. However, velocity drops in moving blades.
- In second set of nozzles, the remaining pressure drop takes place, but velocity increases and, the drop in velocity takes place in moving blades of the second rotor



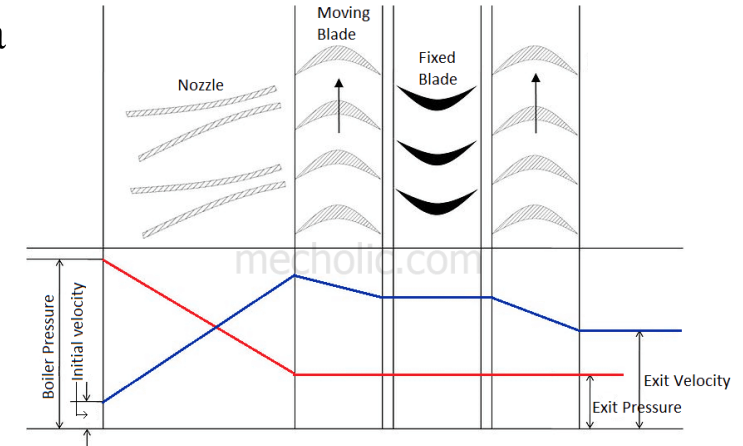
Pressure Velocity Compounding

ADVANTAGES OF COMPOUNDING OF TURBINE

- Compounding reduces the speed of the rotor. If the same energy extracted by the single stage the speed of rotor will too high to practical use.
- Minimize the vibration and noises.
- The turbine compounded for high efficiency with reasonable speed, dimensions.
- The compounding of turbine helps to design the turbine of proper size.
- Reduce the wastage of steam and energy loss due to carry over loss (by the high exit velocity of fluid)

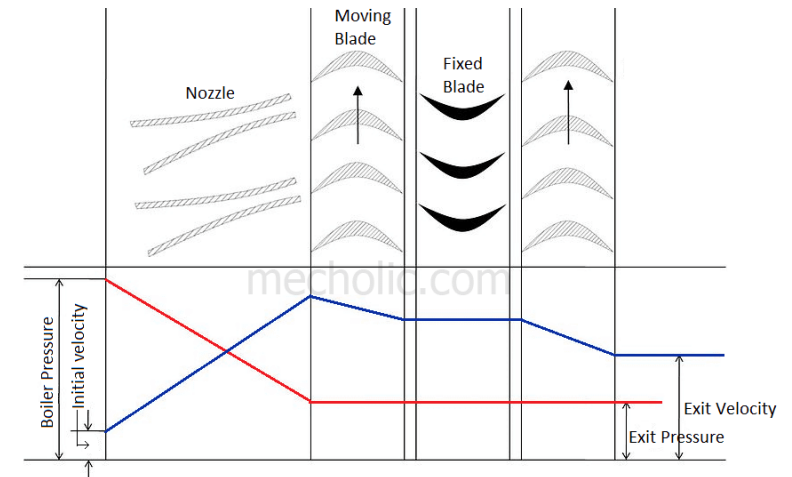
VELOCITY COMPOUNDING

- The turbine has adjacent rows of moving blades and fixed blades.
- The moving blades are keyed to the same shaft. Here the working fluid is a steam.
- Initially, the steam from boiler having high pressure and low velocity is directed to moving blades through a stationary nozzle.
- The steam completely expands in the nozzle, as steam passes through the nozzle the total enthalpy drop and hence the pressure of steam drops.
- At the same time, the velocity or kinetic energy of steam increases. The portion of the kinetic energy is then absorbed by the row of moving blades, impart a rotation on shaft.

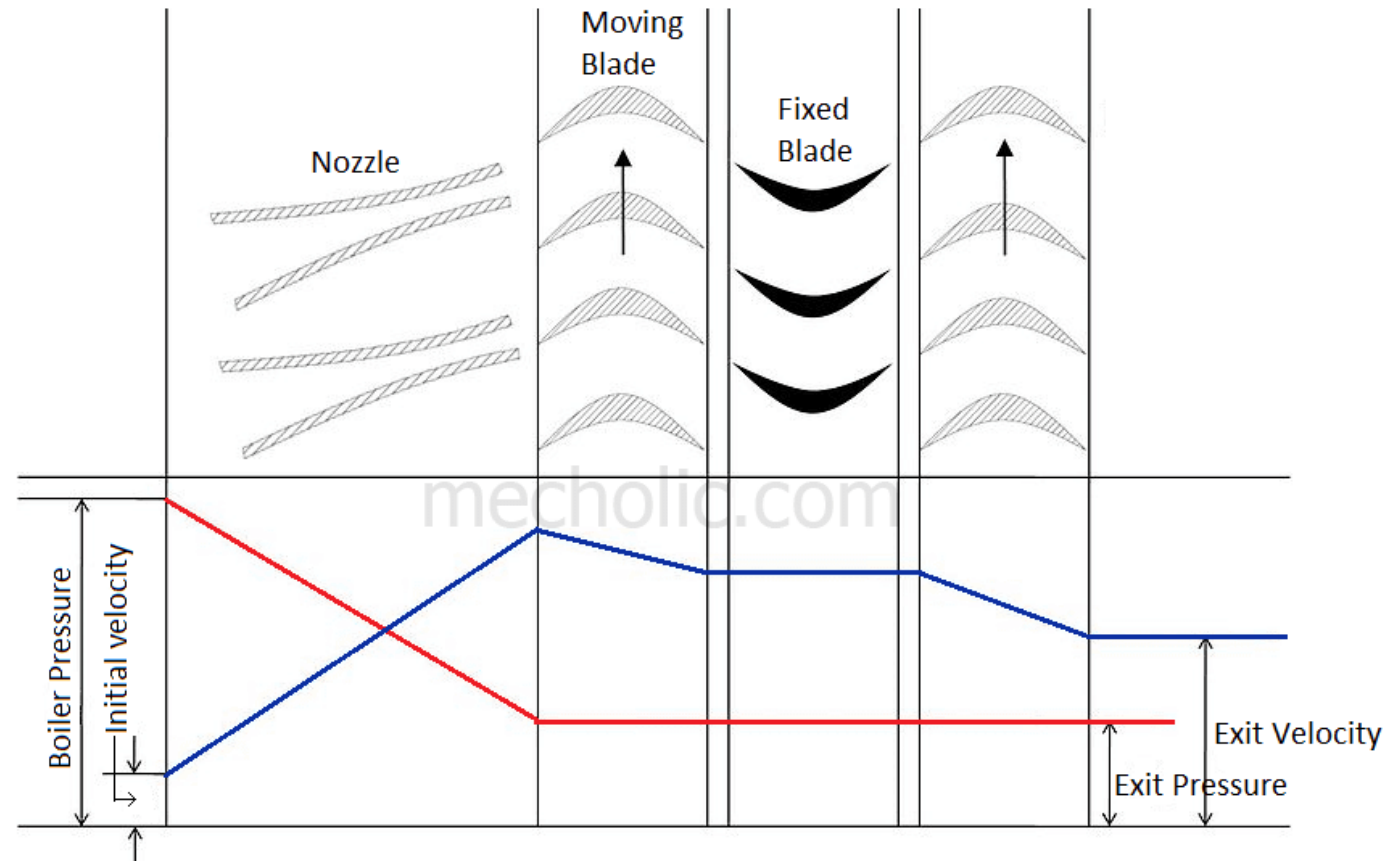


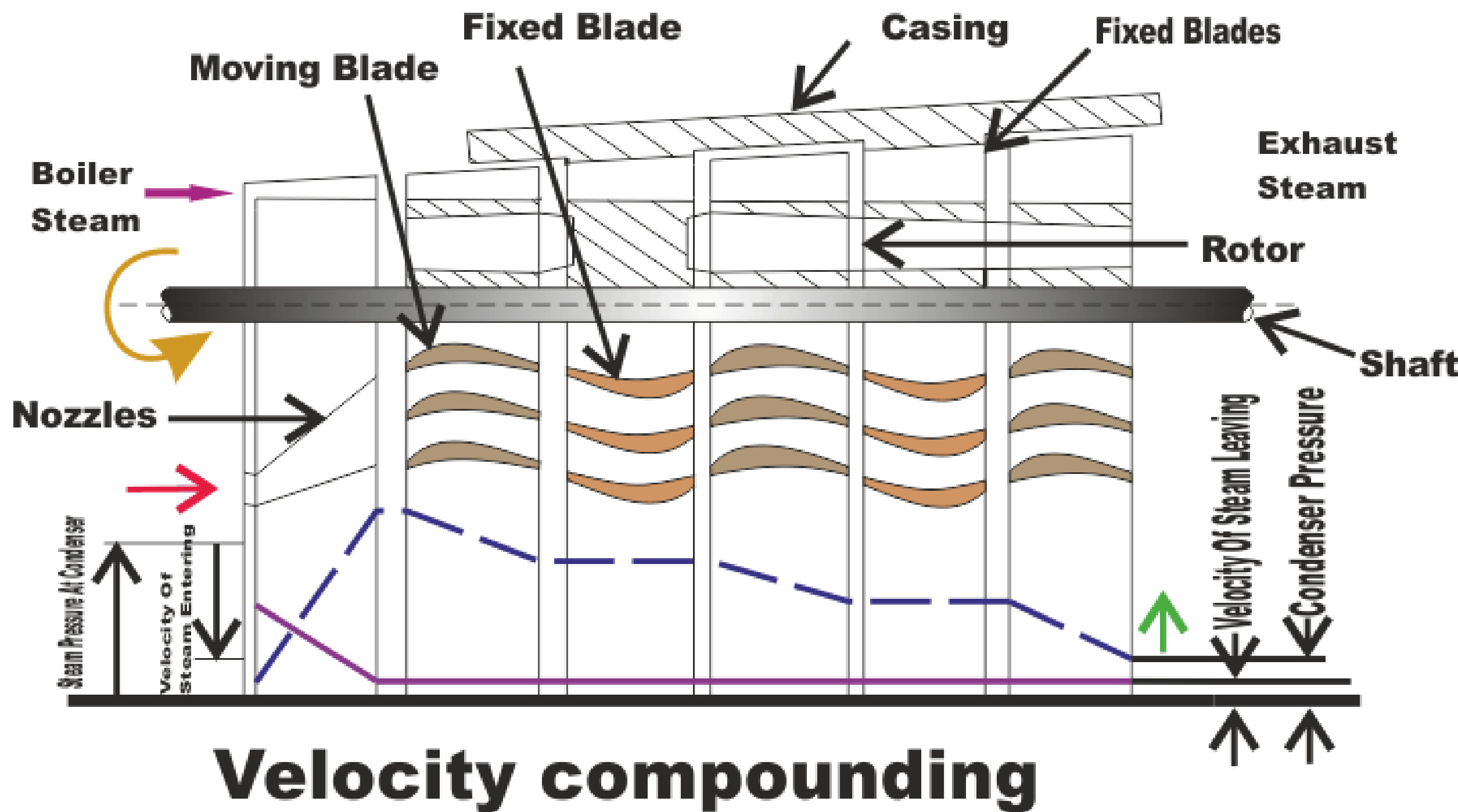
VELOCITY COMPOUNDING

- The velocity of steam decreases and pressure remains the same while it passes over the moving blades.
- The steam then passes through the fixed blades without any change in velocity and the pressure (the velocity will slightly drop in practical application).
- The purpose of the fixed blade is to redirect the steam to the next rows of moving blades where again work is done on the rotor with dissipating kinetic energy.
- Finally, the low-velocity steam leaves the turbine.



VELOCITY COMPOUNDING





VELOCITY COMPOUNDING

Advantages

- Fewer number of stages and length of turbine is short.
- Due to fewer stages, the space required and initial cost is less.
- Low-pressure operation, so the leakage of steam is less.
- Since it works at low-pressure turbine, blades need not be very strong, stress on housing also reduced.
- Reliable and easy to operate.
- Very large heat drop in the first stage. The steam temperature is low at second and third row of wheel, there for cast iron can used for cylinder. This will reduce material cost.

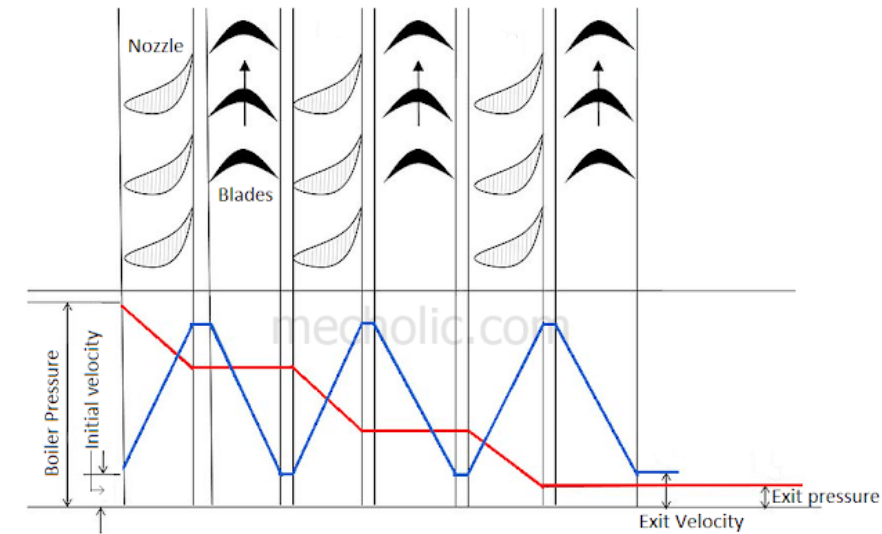
VELOCITY COMPOUNDING

Disadvantages

- Low efficiency because of the ratio of blade velocity to steam velocity not optimum for all wheels.
- High frictional loss due to high velocity.
- High consumption of steam.
- Special designing and fabrication is required for blades which can withstand high velocity.
- First stage produces maximum power, work produced in the low-pressure stages is much less.

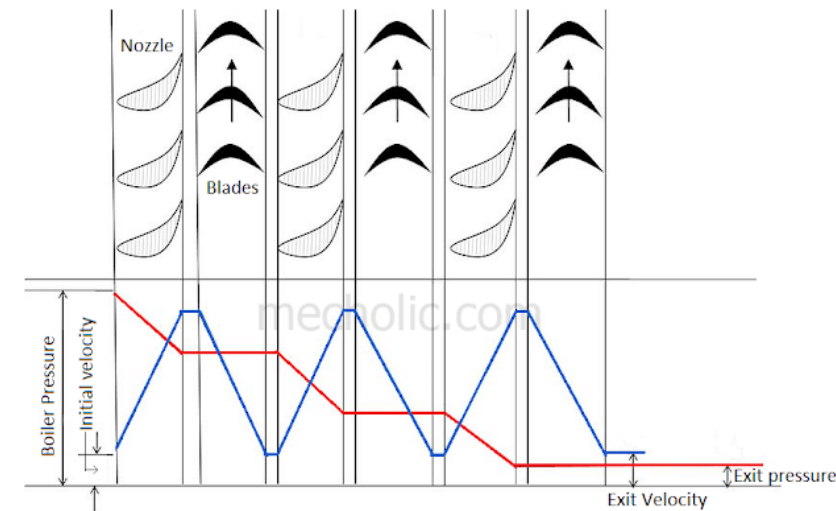
PRESSURE COMPOUNDING

- The pressure compounding is a type of turbine in which the pressure energy of steam is extracted in multiple stages instead of single stages.
- It has rows of moving blades and fixed nozzle arranged adjacently.
- Since it operates at high pressure the whole turbine should be airtight.
- Here the steam is first directed to moving blades through the main nozzle.
- While pass through the main nozzle, steam partially get expanded, the pressure drops, and velocity increase.

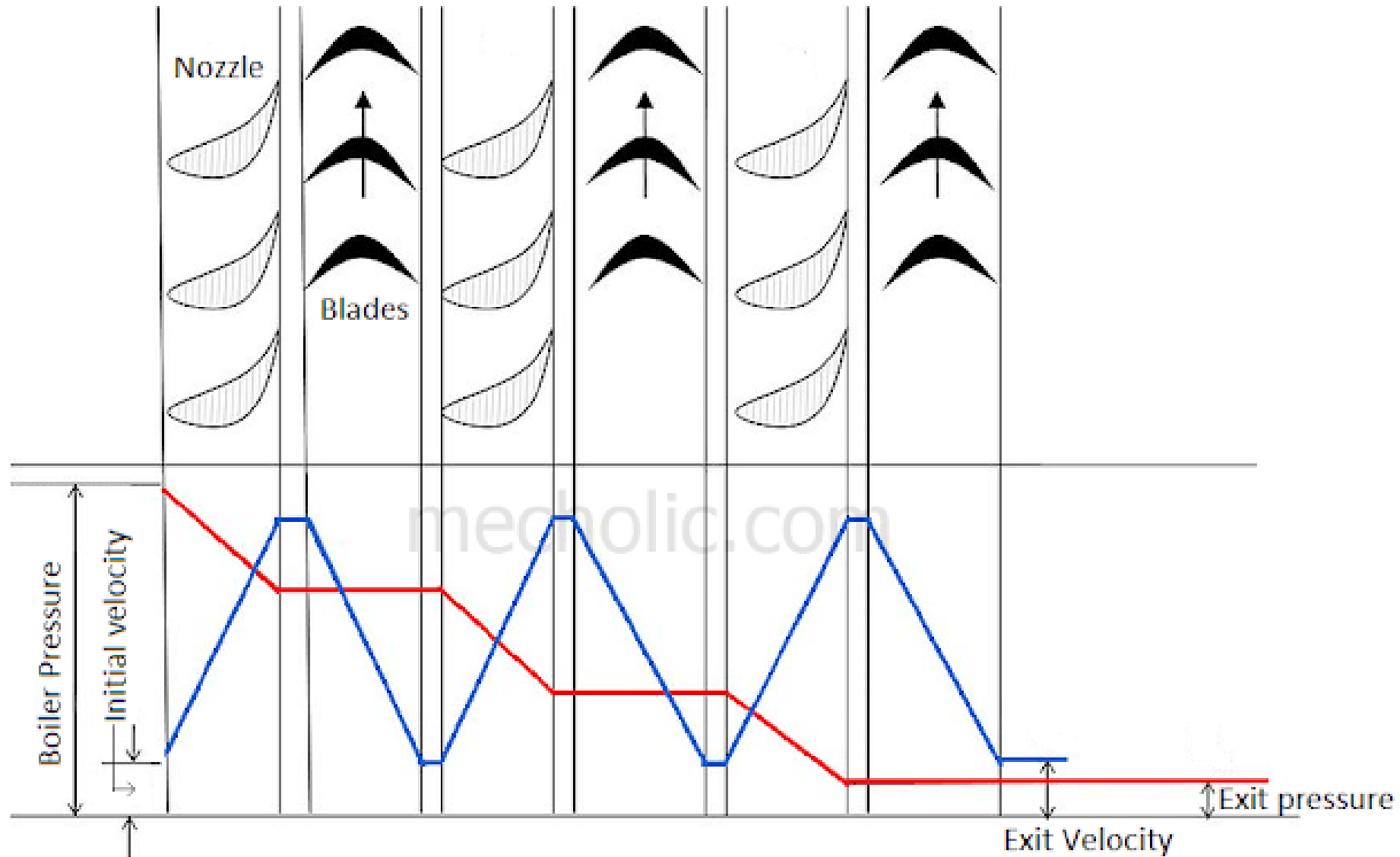


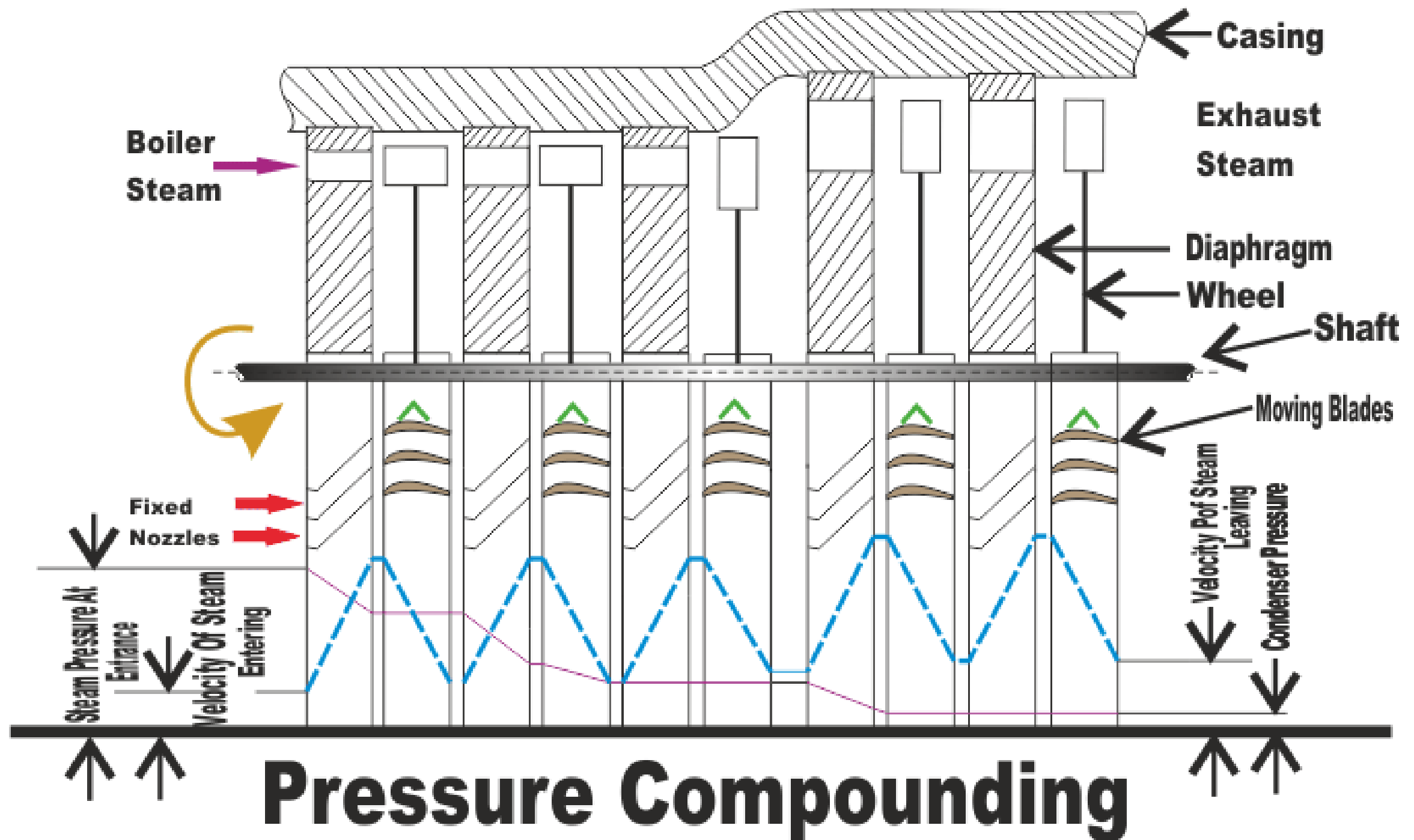
PRESSURE COMPOUNDING

- Approximately all of the gained kinetic energy is then absorbed by the first set moving blades.
- The pressure remains constant while steam passes through the blades.
- The steam then moves through the second set of nozzle, where it again partially get expanded.
- The result will be decreasing pressure and increase velocity.
- Then the steam move to the second set blades, there it continue the similar process happened in first set of blades.



PRESSURE COMPOUNDING





PRESSURE COMPOUNDING

Advantages

- The velocity of steam is lower than that of velocity compound turbine.
- Most efficient turbine because the ratio of blade velocity to steam velocity is constant.

Disadvantages

- Since pressure compounding includes the partial pressure drops in nozzle, the whole turbine should be airtight.
- Large number of stages are required.
- Bulkier than velocity compounding.
- Expensive

PRESSURE-VELOCITY COMPOUNDED IMPULSE TURBINE

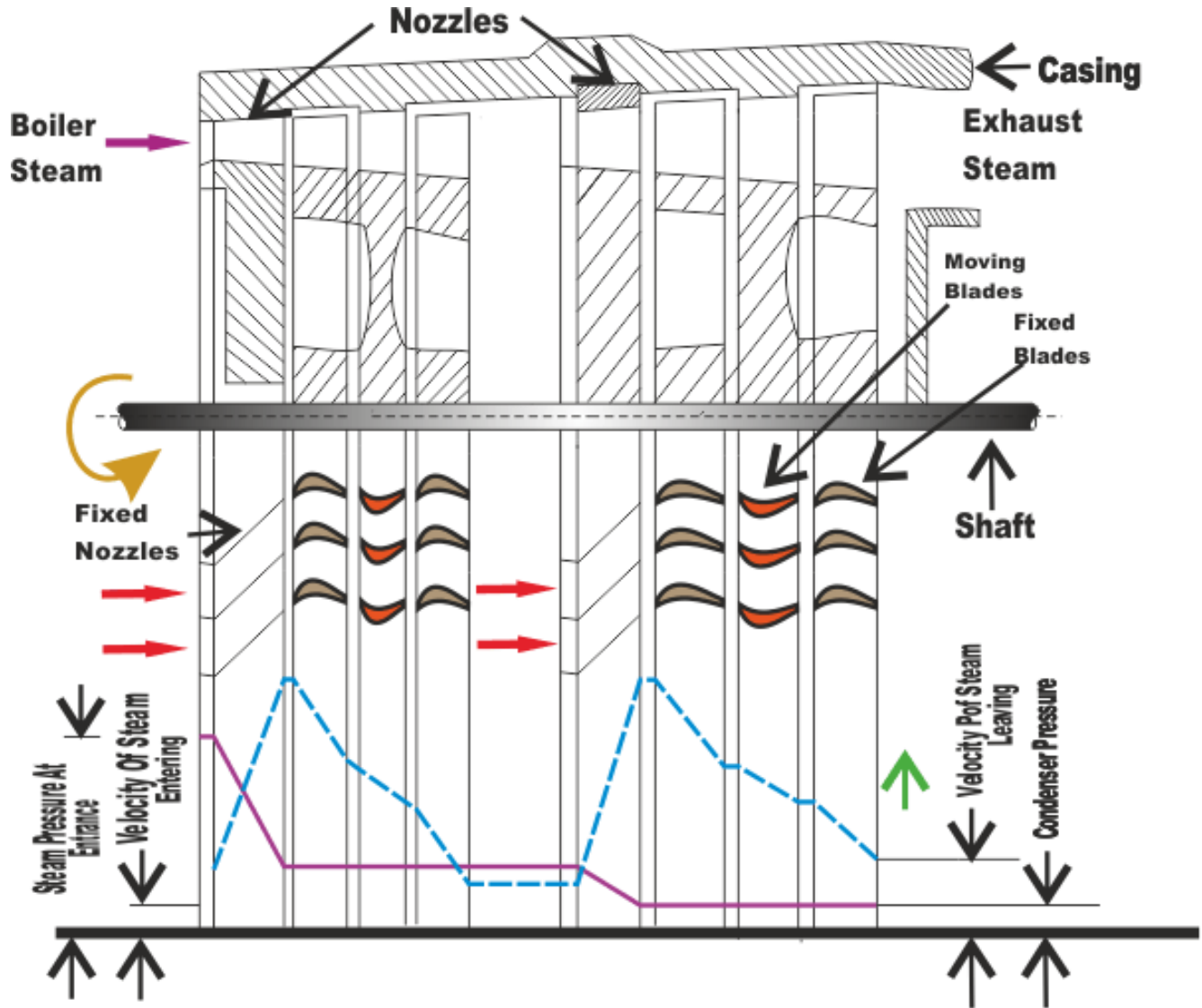
- This turbine is the combination of pressure and velocity compounding.
- It consist of set of nozzles each consist of moving and fixed blades. Pressure compounding occurs in set of nozzles while velocity compounding occurs in moving and fixed blades.

PRESSURE-VELOCITY COMPOUNDED IMPULSE TURBINE

- In first set of nozzles, slight decrease in pressure occurs while the kinetic energy increases. There is no pressure drop in two rows of moving blade. However, velocity drops in moving blades.
- In second set of nozzles, the remaining pressure drop takes place, but velocity increases and, the drop in velocity takes place in moving blades of the second rotor

PRESSURE-VELOCITY COMPOUNDED IMPULSE TURBINE

- In first set of nozzles, slight decrease in pressure occurs while the kinetic energy increases. There is no pressure drop in two rows of moving blade. However, velocity drops in moving blades.
- In second set of nozzles, the remaining pressure drop takes place, but velocity increases and, the drop in velocity takes place in moving blades of the second rotor



Pressure Velocity Compounding

ADVANTAGES OF COMPOUNDING OF TURBINE

- Compounding reduces the speed of the rotor. If the same energy extracted by the single stage the speed of rotor will too high to practical use.
- Minimize the vibration and noises.
- The turbine compounded for high efficiency with reasonable speed, dimensions.
- The compounding of turbine helps to design the turbine of proper size.
- Reduce the wastage of steam and energy loss due to carry over loss (by the high exit velocity of fluid)